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FORMATIVE ASSIGNMENT PART 1: MULTIPLE LINEAR REGRESSION

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1. Introduction

A prominent medical scheme aid in South Africa is looking to develop a model that tailors medical costs according to the lifestyles and geographic regions of their customers. The objective of the analysis is to compute a linear regression model that can estimate medical costs based on demographic, lifestyle and geographic region.

This analysis will use the Medical Cost Personal Dataset (<https://www.kaggle.com/datasets/mirichoi0218/insurance>), which contains information on individuals' age, sex, body mass index (BMI), number of children, smoking habits, geographic region and medical charges.

The objective of this report and analysis is to develop and evaluate a Linear Regression model that will predict medical charges using age, sex, BMI, number of children, smoking status and geographic region. The model will be used to demonstrate how predicted charges can differ between customer profiles and providing a basis for a potential sliding scale approach.

2. Data Suitability

The dataset used has 1338 observations and seven variables including age, sex, BMI, children, smoker, geographic region and charges. The target(dependent) variable is charges which represents the medical costs that is charged per individual. The predictors(independent) variables are age, sex, BMI, children, smoker, geographic region which influence the cost charged per individual.

The dataset is suitable for a linear regression model because the target variable is continuous and numerical. Regression analysis estimates dependent 'y' variable value due to the range of independent variable values 'x' (Maulud and Abdulazeez, 2020). It is suitable for a multiple linear regression model because it has multiple predictor variables. The predictor variables include both numerical and categorical data. The numerical values include age, children and BMI, while the categorical variables include sex, smoker and geographic region. The categorical variables can be converted into numerical representation through one-hot coding, a standard method for converting categorical variables for use in regression models (James et al., 2021) before being used in the regression model.

3. Analysis Planning

The analysis followed the following steps: -

1. Data cleaning and preparation – The dataset was loaded into the notebook and inspected for its dimensions, data types, missing values and duplicate observations. There were no missing values, and the identified duplicate was removed.
2. Exploratory Data Analysis (EDA)- The distributions of the variables and the relationships between the predictors and the medical charges was looked at using descriptive statistics and visualisation.
3. Feature selection- The predictor variables were examined using exploratory findings statistical findings. P-values from an OLS regression were used as evidence when considering the contribution of the predictors. The medical aid scheme's specified factors were also considered when determining the final feature set.
4. Data preparation- The data was divided into training and testing datasets. Categorical variables were converted into numerical features using one hot encoding.

5. Model training - a multiple linear regression model was trained using the prepared predictor variables with charges as the target variable.
6. Model evaluation- The models' predictive performance was evaluated using R-squared(R^2), Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE)
7. Application- The final model was applied to hypothetical customer profiles to demonstrate how predicted medical charges could vary between different individuals.

4. Conducting the analysis

4.1. Exploratory Data Analysis (EDA)

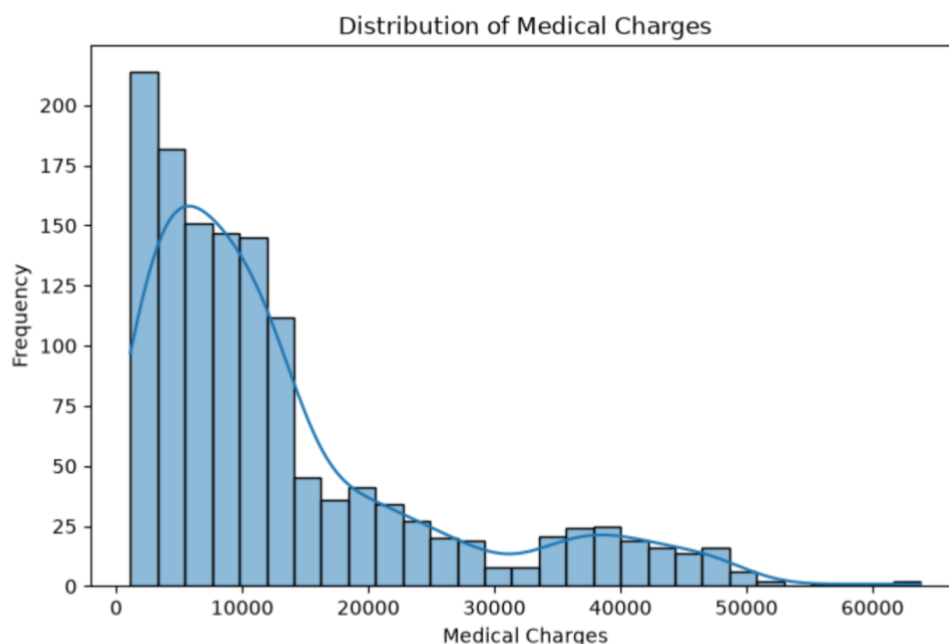
-Overview of the data

The data set was first examined to understand its structure and identify potential data quality issues like missing values and duplicates. The original dataset contained 1338 observations, and seven variables, no missing values were identified but one duplicate observation was found and removed before further analysis leaving the dataset with 1337 observations.

The numerical variables were examined using descriptive statistics to understand their distribution and ranges, while the categorical variables were examined to identify the categories represented in the dataset.

-Distribution of medical charges

the distribution of charges was examined using a histogram the distribution showed that medical charges are not evenly distributed with a concentration of observations having lower charges and a small number of individuals with substantially higher charges.

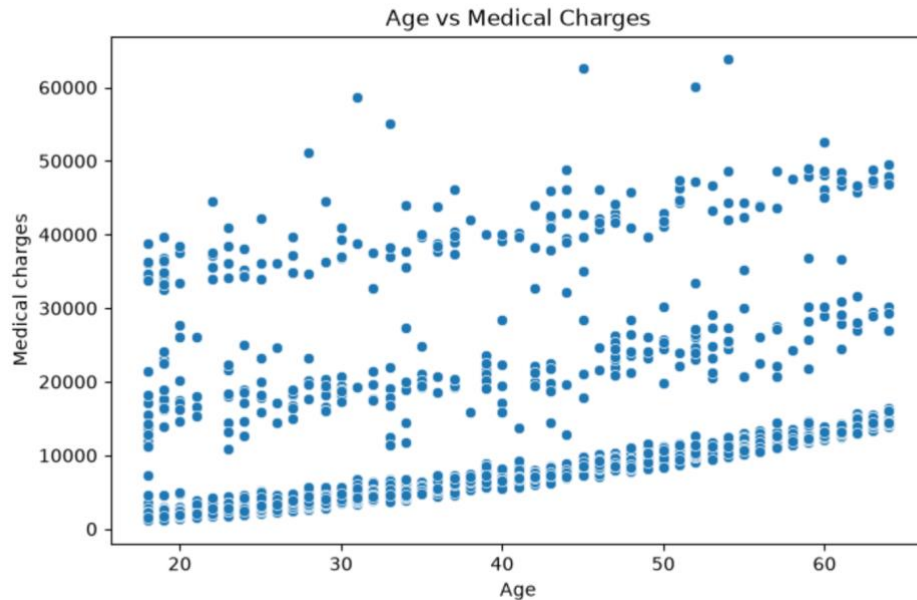


This histogram shows that most medical charges are lower while a small number of charges are much higher. This shows that there is a big difference in medical costs between individuals.

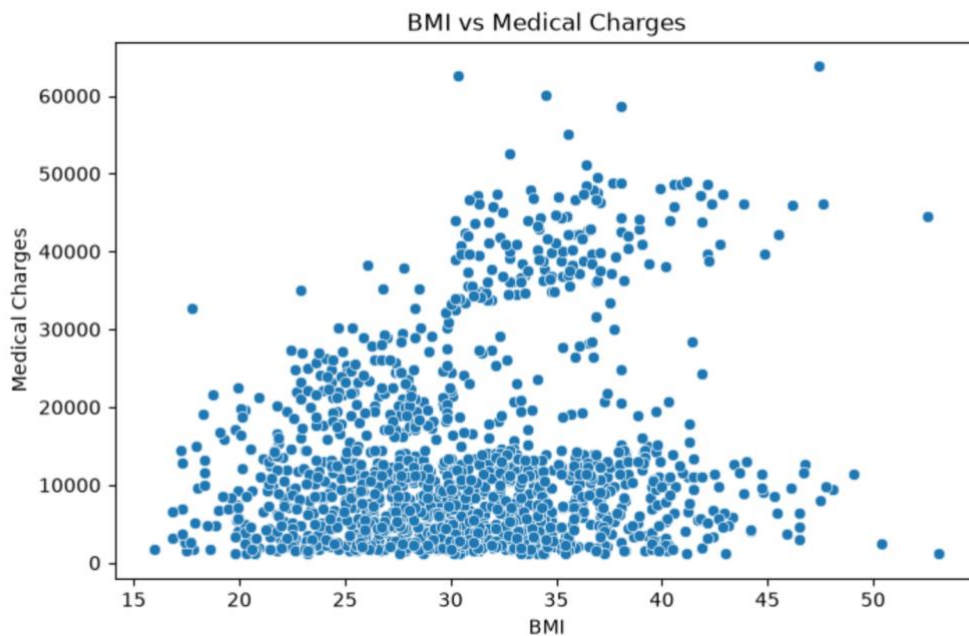
-Relationship between numerical predictors and medical charges

Scatter plots were used to examine the relationships between age, BMI and medical charges.

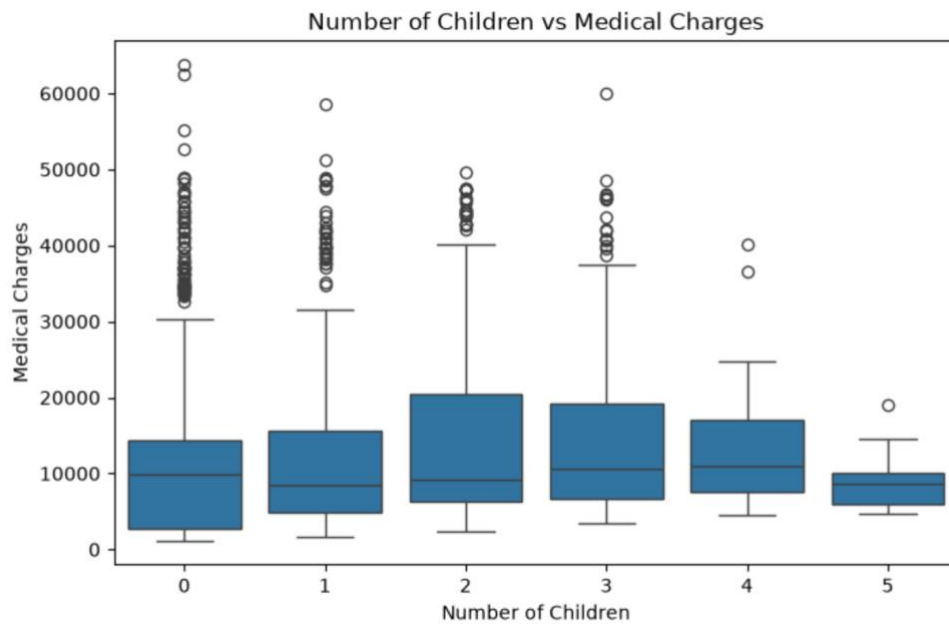
The relationship between age and charges was examined to determine whether medical costs tend to change as individuals become older.



The relationship between BMI and charges was also examined to determine whether BMI has significant influence on an individual's medical charges.



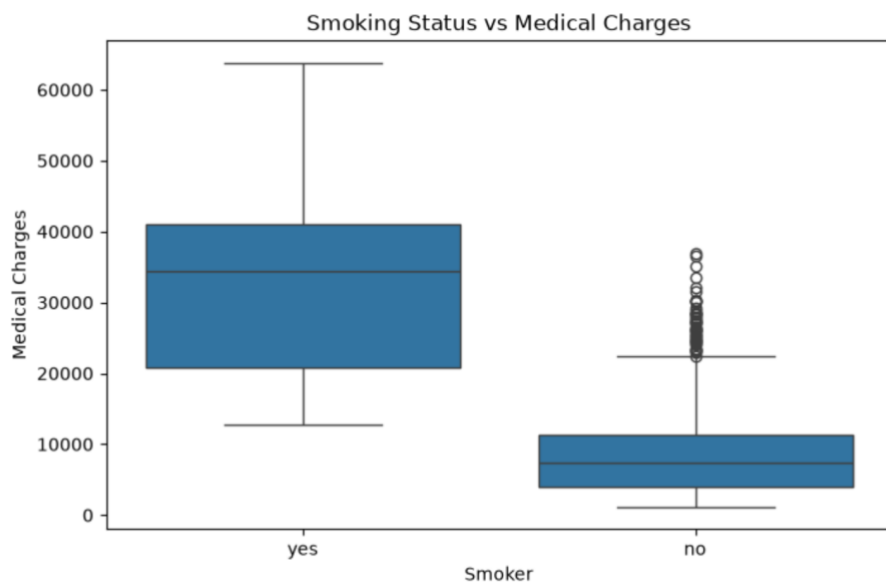
The number of children was similarly examined to determine whether medical charges differed across the different groups.



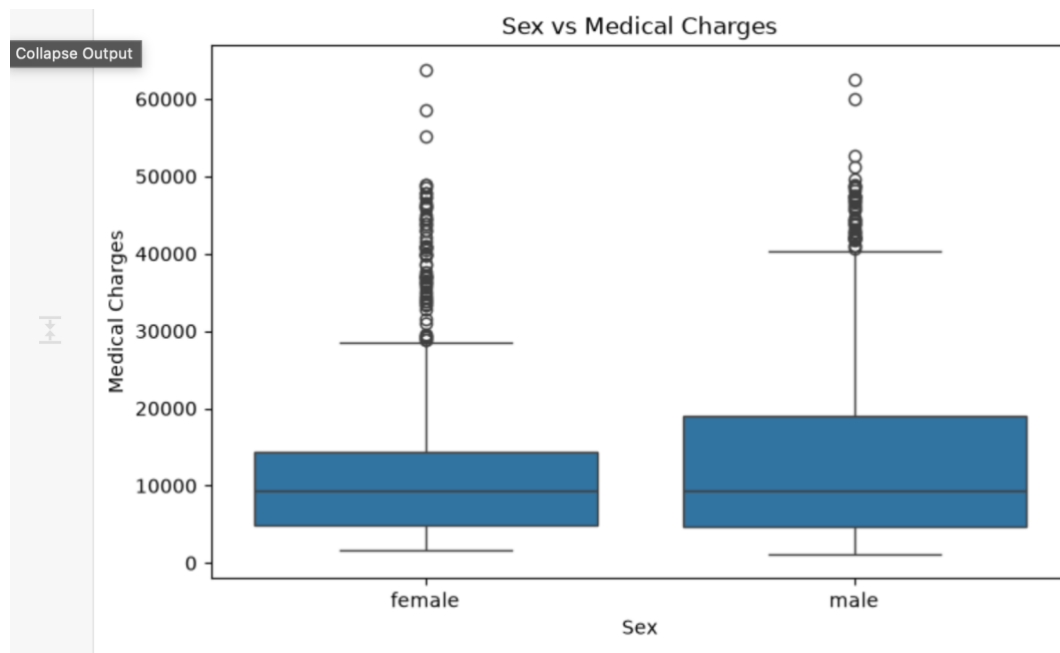
-Categorical variables and charges

Box plots were used to compare medical charges across categories of smoking status, sex, region and number of children

The comparison of smokers and non-smokers was important because substantial differences in medical charges between the groups would indicate that smoking status may be an important predictor.

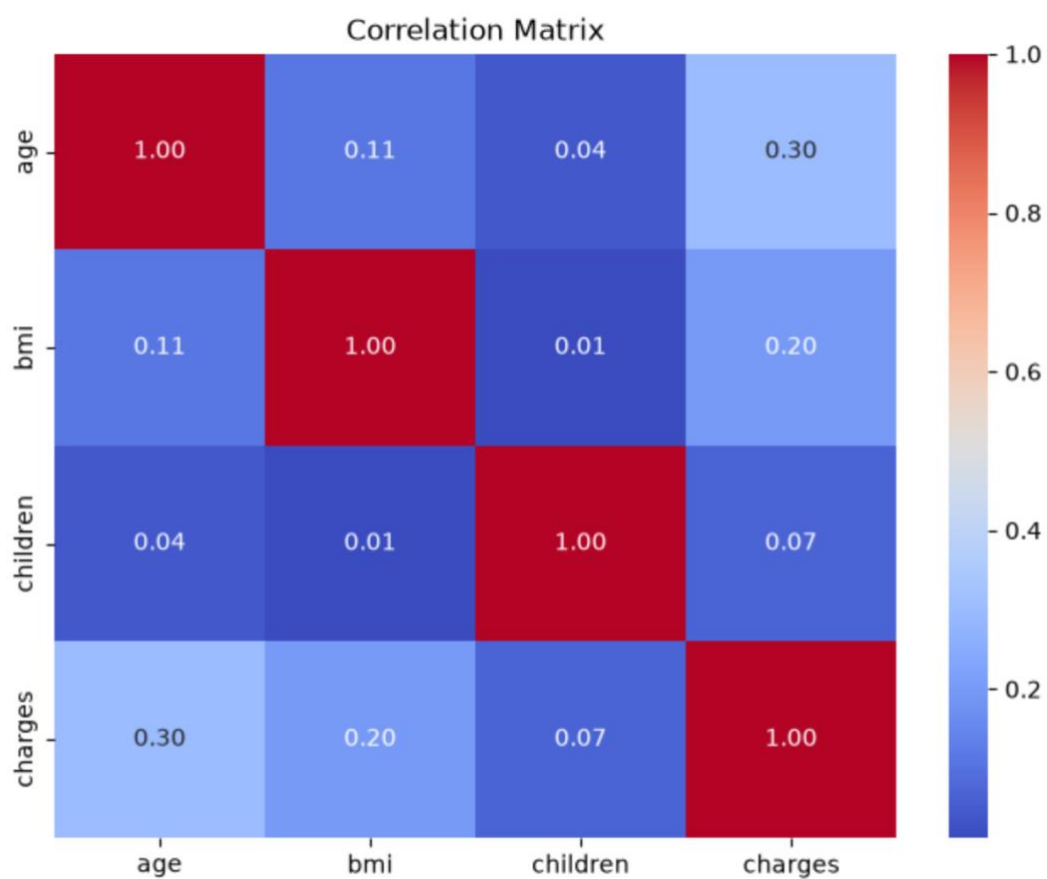


the distribution of charges across sex and region were also examined.



-Correlation Analysis

A correlation matrix was used to examine the relationships between the numerical values.



The correlation analysis provided an initial indication of the strength and direction of relationships between the numerical predictors and charges these findings together with the

categorical comparisons and other EDA findings were used to inform the selection of variables for the regression model.

4.2 Feature selection and Data Preparation

The variable selected for the model were age, sex, BMI, number of children, smoking status and geographic region. These variables were selected because they are the factors specified in the brief and are available in the data set.

The Exploratory Analysis showed that age, BMI and smoking status have a significant relationship with medical charges. The remaining variables were kept because they may contribute differences in charges when considered together with the other predictors.

The resulting prepared data was then used to train the multiple linear regression model.

4.3 Training the model

A multiple linear regression model was trained using the six selected predictors

- Age
- Sex
- BMI
- Number of children
- Smoking status
- Region

The dataset was divided into 80% training data and 20% testing data. Numerical variables were retained in their numerical form while categorical variables were converted into numerical representation using one hot encoding. The default parameters of the linear regression model were used because the objective was to establish a baseline model using the predictors specified by the client.

5. Model Evaluation

The model was evaluated on the test data set using R-squared, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

R²- the model got a R² of 0.8069

This shows that about 80.7% of the variation in medical charges in the test data set can be explained by the predictors included in the model.

MAE- the model got a mean absolute error of 4177.05

This shows that on average the models predicted medical charge differs from the actual charge by about R4177 charge on average

RMSE- the model got an RMSE of 5956.34

“RMSE gives greater weight to large prediction errors than MAE, this makes it useful for determining whether a model occasionally produces substantially inaccurate predictions (Chai and Draxler, 2014).”

Based on these results, the model provides a reasonable baseline for estimating medical charges or has limitations in accurately predicting individual charges. The difference between

the MAE and the RMSE shows that the model is less accurate for a subset of cases with very high charges.

6. Model Retraining

The model gave reasonable results with an R square of 0.8069. Although some prediction errors were pre or present the model explained about 80.7% of the variation in medical charges therefore retraining was considered not necessary for this model.

Sliding scale

The trained model was used to predict medical charges for three hypothetical customer profiles with different demographic lifestyle and geographic characteristics the predicted values demonstrate how the model can be used to estimate different charges for different customer profiles providing a basis for a sliding scale approach.

	age	sex	bmi	children	smoker	region	predicted_charges
0	25	female	22.5	0	no	southeast	1444.478515
1	40	male	28.0	2	no	northwest	8332.133329
2	55	female	32.0	3	yes	southwest	36775.038235

These results show the sliding scale effect because smoking status can result in a charge that is more than a charge of a non-smoking profile or individual.

6. Conclusion

A multiple linear regression model was developed to predict medical charges by using age, sex, BMI, number of children, smoking status and region. The data was cleaned and explored before the variables were prepared and the model was trained and evaluated using R-squared, MAE and RMSE the model was also used to demonstrate how predicted charges could vary between different customer profiles.

7. References

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